Question ID b52e5b6f

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in two variables	■□□

ID: b52e5b6f

PSAT2 M 1.1

A mixture consisting of only vitamin D and calcium has a total mass of 150 grams. The mass of vitamin D in the mixture is 50 grams. What is the mass, in grams, of calcium in the mixture?

A. 200

B. 150

C. 100

D. 50

Question ID fa80893a

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in one variable	■□□

ID: fa80893a

PSAT2 M 1.2

If $2x + 3 = 9$, what is the value of $6x - 1$?

Question ID a478f9f5

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Probability and conditional probability	■□□

ID: a478f9f5

PSAT2 M 1.3

Each of 157 gemstones can be classified as one of three classifications, as shown in the frequency table.

Classification	Frequency
color X	119
color Y	3
color Z	35

If one of the gemstones is selected at random, what is the probability of selecting a gemstone of color Y?

A. $\frac{3}{157}$

B. $\frac{35}{157}$

C. $\frac{119}{157}$

D. $\frac{154}{157}$

Question ID ed856e9c

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in two variables	■□□

ID: ed856e9c

PSAT2 M 1.4

What is the *y*-intercept of the graph of $y = 34x + 81$ in the *xy*-plane?

A. $(0, 81)$

B. $(0, 34)$

C. $(0, -34)$

D. $(0, -81)$

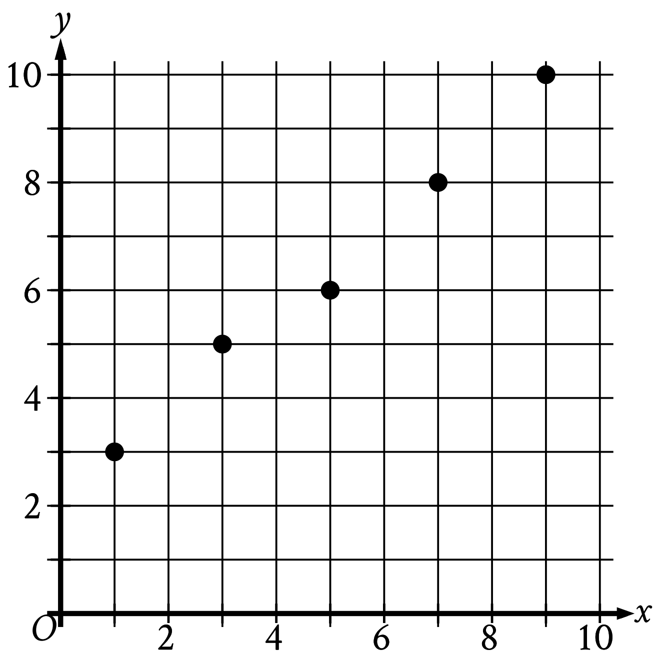
Question ID 16988f9c

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Two-variable data: Models and scatterplots	■□□

ID: 16988f9c

PSAT2 M 1.5

The scatterplot shows the relationship between two variables, x and y.



Which equation is the most appropriate linear model for this relationship?

A. $y = -0.9x - 2.2$

B. $y = -0.9x + 2.2$

C. $y = -0.9x$

D. $y = 0.9x + 2.2$

Question ID 0700a2d5

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Ratios, rates, proportional relationships, and units	■□□

ID: 0700a2d5

PSAT2 M 1.6

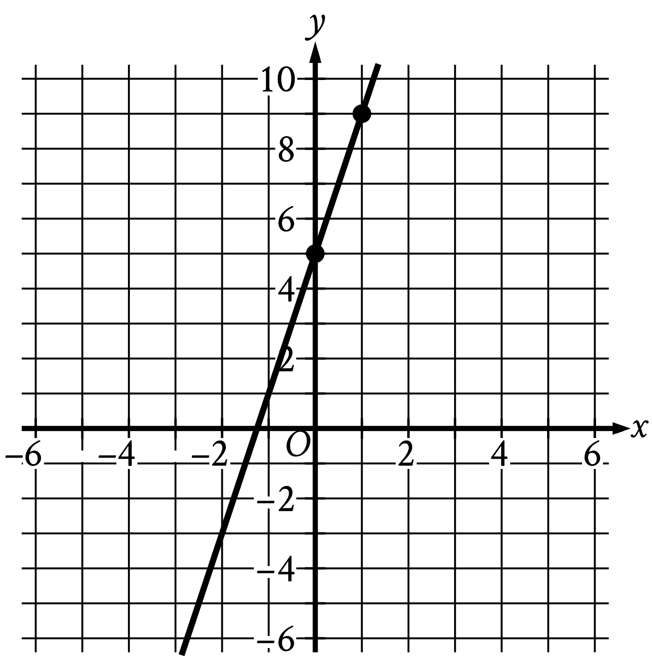
How many yards are equivalent to 77 rods? $(5.5 \text{ yards} = 1 \text{ rod})$

Question ID b8fa27db

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in two variables	■□□

ID: b8fa27db

PSAT2 M 1.7



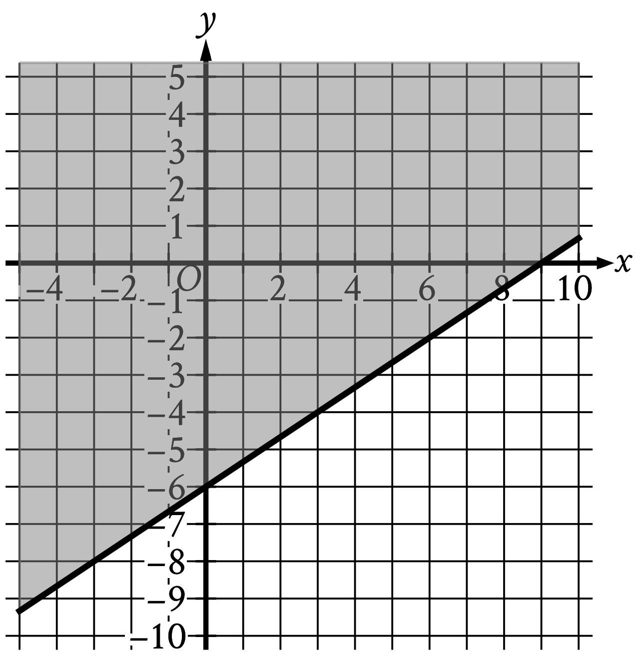
Line j is shown in the *xy*-plane. Line k (not shown) is parallel to line j. What is the slope of line k?

Question ID c9355dec

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear inequalities in one or two variables	■□□

ID: c9355dec

PSAT2 M 1.8



The shaded region shown represents the solutions to which inequality?

A. $y \geq \frac{2}{3}x - 6$

B. $y \geq \frac{2}{3}x + 6$

C. $y \geq \frac{2}{3}x - 9$

D. $y \geq \frac{2}{3}x + 9$

Question ID e86f0651

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	■■□

ID: e86f0651

PSAT2 M 1.9

A circle has a radius of 43 meters. What is the area, in square meters, of the circle?

A. $\frac{43\pi}{2}$

B. 43π

C. 86π

D. $1{,}849\pi$

Question ID 3a3b95df

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	■□□

ID: 3a3b95df

PSAT2 M 1.10

$d = 16 - \frac{x}{30}$

The equation shown gives the estimated amount of diesel d, in gallons, that remains in the gas tank of a truck after being driven x miles, where $0 \leq x \leq 480$. What is the estimated amount of diesel, in gallons, that remains in the gas tank of the truck when $x = 300$?

A. 0

B. 6

C. 14

D. 16

Question ID 739f1bbc

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	■□□

ID: 739f1bbc

PSAT2 M 1.11

In triangle ABC, $AB = 4{,}680 \text{ millimeters } (\text{mm})$ and $BC = 4{,}680 \text{ mm}$. Which statement is sufficient to prove that triangle ABC is equilateral?

A. $AC = 4{,}680 \text{ mm}$

B. $AC = 468 \text{ mm}$

C. $AC = 46.8 \text{ mm}$

D. $AC = 4.68 \text{ mm}$

Question ID 07bcecac

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	■□□

ID: 07bcecac

PSAT2 M 1.12

$P(t) = 24.8(1.036)^t$

The function P gives the predicted population, in millions, of a certain country for the period from 1984 to 2018, where t is the number of years after 1984. According to the model, what is the best interpretation of the statement "$P(8)$ is approximately equal to 32.91"?

A. In 1984, the predicted population of this country was approximately 8 million.

B. In 1984, the predicted population of this country was approximately 32.91 million.

C. 8 years after 1984, the predicted population of this country was approximately 32.91 million.

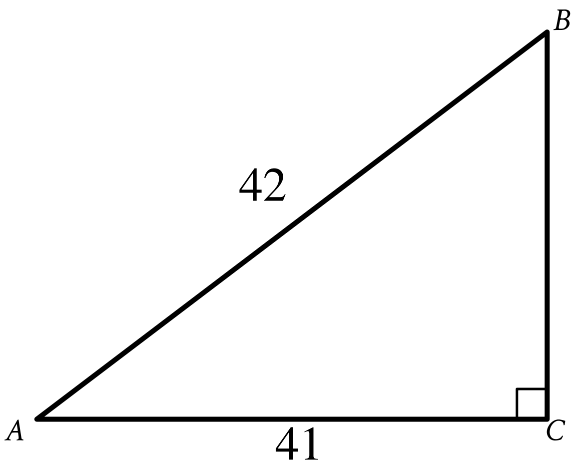
D. 32.91 years after 1984, the predicted population of this country was approximately 8 million.

Question ID 2bddbc1b

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Right triangles and trigonometry	■■□

ID: 2bddbc1b

PSAT2 M 1.13



Note: Figure not drawn to scale.

What is the value of $\cos A$ in the triangle shown?

A. $\frac{42}{41}$

B. $\frac{41}{42}$

C. $\frac{1}{42}$

D. $\frac{1}{41}$

Question ID 79cf8505

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in one variable	■■□

ID: 79cf8505

PSAT2 M 1.14

A line segment that has a length of 115 centimeters (cm) is divided into three parts. One part is $47\ \mathrm{cm}$ long. The other two parts have lengths that are equal to each other. What is the length, in cm, of one of the other two parts of equal length?

Question ID a29e89fc

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	One-variable data: Distributions and measures of center and spread	■■□

ID: a29e89fc

PSAT2 M 1.15

The list gives the mass, in grams, of 5 alpine marmots. $4{,}010$; $4{,}010$; $3{,}030$; $4{,}050$; $3{,}050$

What is the mean mass, in grams, of these 5 alpine marmots?

Question ID b4acba95

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	■■□

ID: b4acba95

PSAT2 M 1.16

$x^2 - 12x + 27 = 0$ How many distinct real solutions does the given equation have?

A. Exactly two

B. Exactly one

C. Zero

D. Infinitely many

Question ID 946ab892

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	■■□

ID: 946ab892

PSAT2 M 1.17

For the linear function g, the graph of $y = g(x)$ in the *xy*-plane has a slope of 2 and passes through the point $(1, 14)$. Which equation defines g?

A. $g(x) = 2x$

B. $g(x) = 2x + 2$

C. $g(x) = 2x + 12$

D. $g(x) = 2x + 14$

Question ID 1b403590

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Ratios, rates, proportional relationships, and units	■■□

ID: 1b403590

PSAT2 M 1.18

An object has a mass of 168 grams and a volume of 24 cubic centimeters. What is the density, in grams per cubic centimeter, of the object?

A. 7

B. 144

C. 192

D. $4,032$

Question ID 02add2d2

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	■■□

ID: 02add2d2

PSAT2 M 1.19

A company has a newsletter. In January 2018, there were $1{,}300$ customers subscribed to the newsletter. For the next 24 months after January 2018, the total number of customers subscribed to the newsletter each month was 7% greater than the total number subscribed the previous month. Which equation gives the total number of customers, c, subscribed to the company's newsletter m months after January 2018, where $m \leq 24$?

A. $c = 1{,}300^{(0.07)^m}$

B. $c = 1{,}300^{(1.7)^m}$

C. $c = 1{,}300^{(1.07)^m}$

D. $c = 1{,}300^{(7)^m}$

Question ID 39714777

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	■■□

ID: 39714777

PSAT2 M 1.20

$p(x) + 57 = x^2$

The given equation relates the value of x and its corresponding value of $p(x)$ for the function p. What is the minimum value of the function p?

A. $-3{,}249$

B. -57

C. 57

D. $3{,}249$

Question ID 1baffbcf

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Percentages	■■■

ID: 1baffbcf

PSAT2 M 1.21

The expression $0.35x$ represents the result of decreasing a positive quantity x by what percent?

A. 3.5%

B. 35%

C. 6.5%

D. 65%

Question ID 7866a908

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Systems of two linear equations in two variables	■■■

ID: 7866a908

PSAT2 M 1.22

A sample of a certain alloy has a total mass of 50.0 grams and is 50.0% silicon by mass. The sample was created by combining two pieces of different alloys. The first piece was 30.0% silicon by mass and the second piece was 80.0% silicon by mass. What was the mass, in grams, of the silicon in the second piece?

A. 9.0

B. 16.0

C. 20.0

D. 30.0